

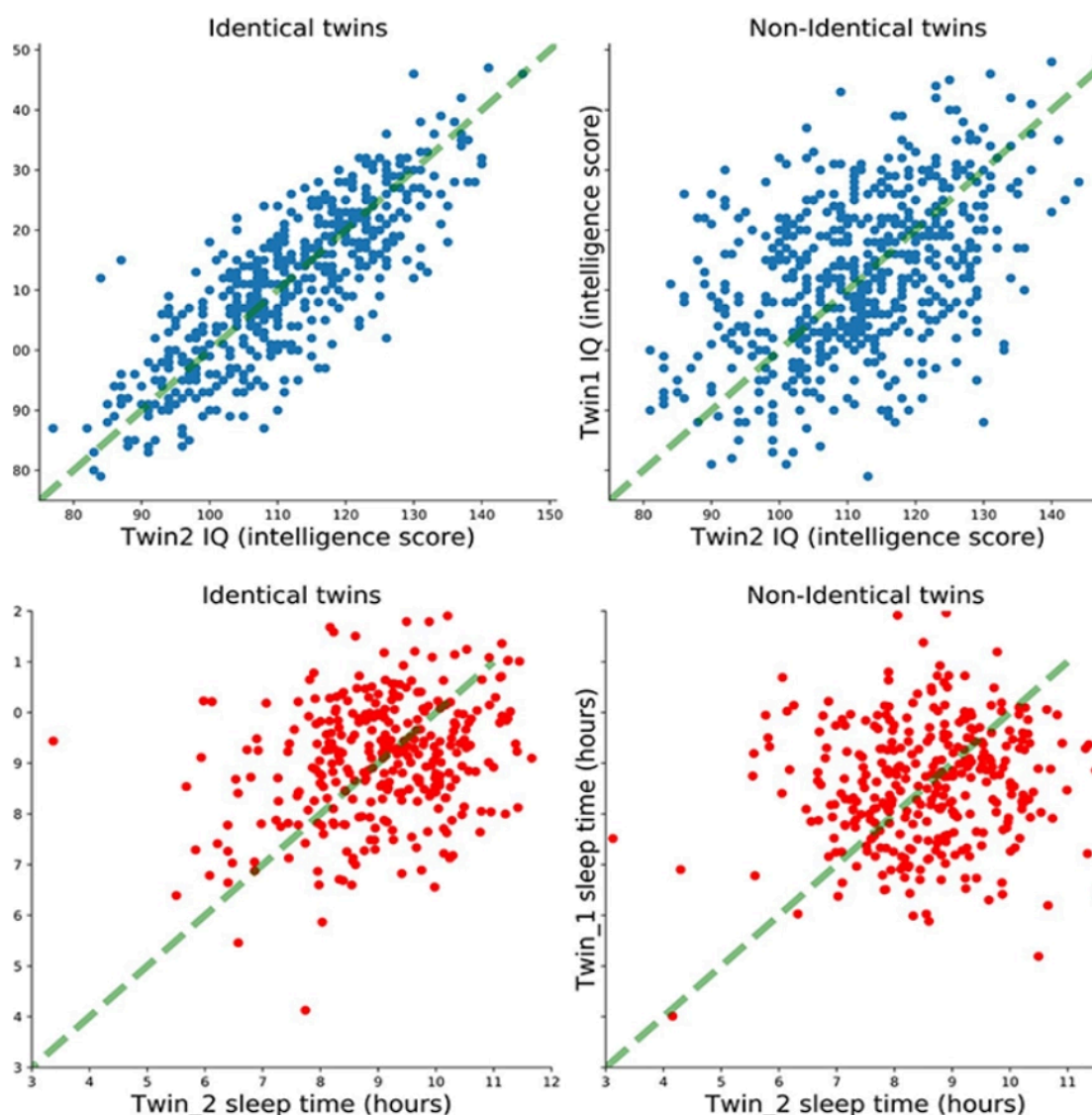
5.3 Complex Disease and Pharmacogenetics

- ▼ What is a Mendelian trait?
 - A trait that is **controlled by a single gene**. Put simply, it is not a complex trait.
 - i.e eye colour
- ▼ What is a complex trait?
 - A trait that is **controlled by multiple genes** + the effect of the **environment**.
 - i.e height, weight, intelligence, blood pressure.
- ▼ Why is sickle cell disease thought to be more prevalent amongst those of Afro-Caribbean descent?
 - The **sickle cell phenotype** was thought to have given protection against malaria.
- ▼ What does the term heritability describe?
 - It describes the **level of contribution that genetics has to any one given phenotype**.
 - i.e 0% genetic to 100% genetic
 - Most phenotypes fall somewhere in between this scale.
- ▼ What is the difference between monozygotic and dizygotic twins?
 - **Monozygotic twins** arise from the fertilisation of one egg by one sperm, and this zygote splitting into two
 - These are identical twins as they have the same genomes.
 - **Dizygotic twins** arise from the fertilisation of two eggs by two sperms, resulting in the formation of two zygotes
 - It is effectively the same as having **two siblings who are born at the same time** instead of a couple of years apart.
 - These twins are described as fraternal twins.

▼ How are twin studies used to estimate heritability?

- Phenotypes are observed in both monozygotic and dizygotic twins
- The **difference in phenotypic expression between monozygotic twins** (who have the same genome) **as well as dizygotic twins** is observed and then we can estimate heritability
 - i.e if a trait is identical in monozygotic but not in dizygotic twins, then we can estimate that heritability of this trait is pretty high

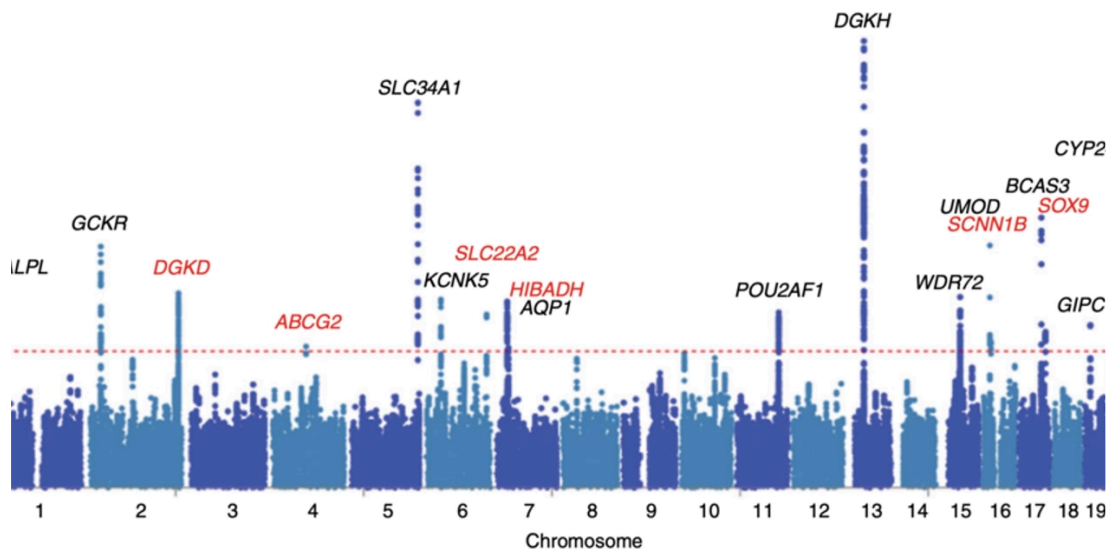
▼ Diagram showing how the heritability of different traits are measured using twin studies.



- ▼ Which big underlying assumption may undermine the validity of twin study results?
 - That the **environmental differences between the two sets of twins are identical**
 - There are so many environmental differences between the two sets of twins that it is almost impossible to account for this in the studies.
- ▼ What is the most common type of mutation in the human genome?
 - **Single Nucleotide Polymorphisms (SNPs)**
 - Each SNP represents a difference in a single DNA building block, called a nucleotide
- ▼ How do association studies work?
 - They work by analysing the **concordance between a particular SNP (mutation) and risk of a disease**
 - A group of **controls** (people without the disease) and **sufferers** are assembled
 - We then look for the **level of correlation** between the type of SNPs present in the population and **how frequently these SNPs cause disease phenotypes**.
- ▼ Describe how the genetic causes of aggression were studied using association studies.
 - There was an **X-linked mutation** noticed in the **gene coding for monoamine oxidase (MAOA)**
 - This mutation led to **monoamines** such as adrenaline, noradrenaline, serotonin, dopamine **not being metabolised**
 - The presence of the mutation was associated with aggressive behaviour in carriers of the mutation
- ▼ Why are association studies limited?
 - Because there is such a great focus on specific mutations having specific effects, **we may miss out something else that could be a causative factor**
 - There is also **significant bias to causative factors that we know of** (i.e there may be a causative factor that science hasn't discovered yet, that is actually responsible for phenotypic changes)
 - **Correlation does not equal causation**

▼ What are genome-wide association studies (GWAS)?

- It is a hypothesis-free study used to identify as many SNPs as possible within a wider section of the genome
- i.e there is no hypothesis (X causes Y), but we are simply trying to find all the different SNPs in the genome which may suggest causation



▼ Recall two benefits of GWAS over association studies.

- They **aren't limited to what we know** - we **may discover more biological mechanisms**
- We can identify individuals at risk of certain conditions due to the presence of particular SNPs.

▼ Recall two limitations of GWAS.

- It doesn't identify causal relationships between SNPs and phenotypes.
- It **doesn't account for environmental changes to the genome (epigenetics)**
- It can be **hard to predict heritability**.
- The threshold set may also mean we miss out on identifying very rare SNPs

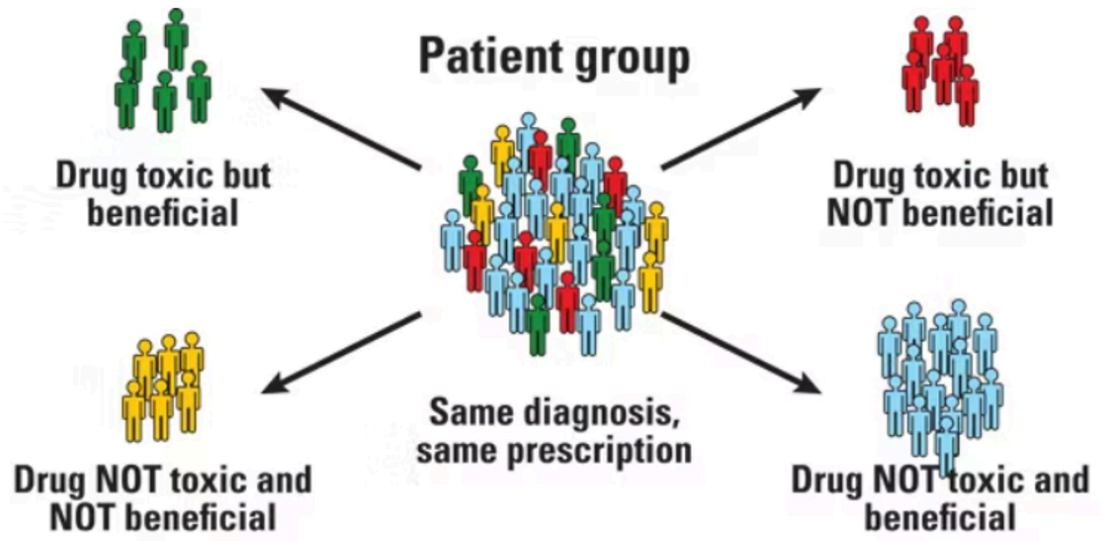
▼ What can whole genome sequencing (WGS) give us over GWAS?

- WGS allows us to analyse the **entire genome** which helps us to **identify every SNP**

▼ What does the term pharmacogenetics refer to?

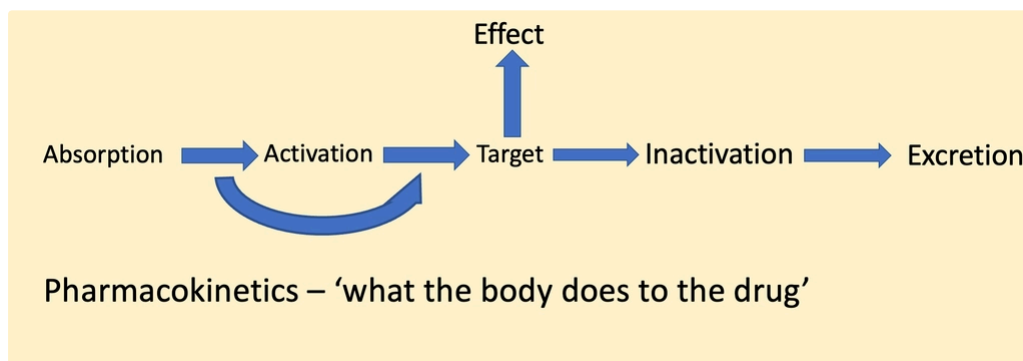
- The study of **variability in responses to drugs as a result of genetic differences**

- ▼ Image showing how drugs can affect people in different ways.



- ▼ What does the term pharmacokinetics refer to?

- It describes the **how the body acts on a particular drug**



- ▼ Describe how mutations in proteins involved in drug absorption can lead to adverse effects.

- It can lead to **unusually high or low levels of the drug accumulating in the blood** which means that doses that are usually of the right level in normal people, become all of a sudden too high or low in these individuals.
 - The effects can be mitigated by administering higher or lower doses of said drug.
 - i.e mutations in absorption proteins in the duodenum affecting drug absorption
- SNPs can lead to changes in enzyme activity which can **alter the activation/inactivation of different drugs** after consumption

▼ What is a prodrug?

- A **drug that doesn't actually have a direct effect on the user itself**, but is instead **metabolised to another substance which can have the desired effect**.
- i.e codeine is a prodrug that is metabolised to morphine.

▼ Why is codeine used instead of morphine directly?

- Codeine has **less depressive and sedative side effects**